

# 6.1–6.2 Eigenvalues, Eigenvectors, and Diagonalization

*Concise exam revision notes for eigenvectors, characteristic equations, powers, and stability.*

Based on handwritten MIT 18.06 revision notes.

## 1 Core Idea

**Core idea.** Most vectors change direction when multiplied by  $A$ . Eigenvectors keep their direction; only their length and sign are scaled.

The defining equation is

$$Ax = \lambda x, \quad x \neq 0.$$

Equivalently,

$$(A - \lambda I)x = 0.$$

So  $A - \lambda I$  must be singular:

$$\det(A - \lambda I) = 0.$$

## 2 How to Find Eigenvalues and Eigenvectors

1. Solve  $\det(A - \lambda I) = 0$  to find eigenvalues.
2. For each eigenvalue, solve  $(A - \lambda I)x = 0$ .
3. Use any nonzero vector in that nullspace as an eigenvector.

**Exam trigger.** Eigenvectors come from a nullspace. If the nullspace has more than one direction, choose a convenient basis.

## 3 Fast Checks

| Check         | Formula                                            | Meaning                         |
|---------------|----------------------------------------------------|---------------------------------|
| Trace         | $\text{trace}(A) = \lambda_1 + \cdots + \lambda_n$ | Sum of eigenvalues.             |
| Determinant   | $\det A = \lambda_1 \lambda_2 \cdots \lambda_n$    | Product of eigenvalues.         |
| Invertibility | 0 is not an eigenvalue                             | Equivalent to $\det A \neq 0$ . |

These checks catch arithmetic mistakes after solving the characteristic equation.

## 4 Special Matrices

| Matrix type           | Typical eigenvalues | Exam memory                                   |
|-----------------------|---------------------|-----------------------------------------------|
| Projection $P$        | 0 and 1             | $P^2 = P$ .                                   |
| Reflection $R$        | 1 and $-1$          | Directions along and across the mirror.       |
| Rotation in the plane | Usually complex     | Real rotations may have no real eigenvectors. |
| Triangular matrix     | Diagonal entries    | Read eigenvalues from the diagonal.           |

**Common mistake.** Row reduction changes eigenvalues in general. Do not row-reduce  $A$  and read eigenvalues from the new matrix.

## 5 Diagonalization

If  $A$  has  $n$  independent eigenvectors, put them into

$$S = \begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix}.$$

Then

$$AS = S\Lambda, \quad A = S\Lambda S^{-1}.$$

Here

$$\Lambda = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix}.$$

| Case                     | Diagonalization result                         |
|--------------------------|------------------------------------------------|
| $n$ distinct eigenvalues | Automatically enough independent eigenvectors. |
| Repeated eigenvalue      | May or may not have enough eigenvectors.       |
| Not enough eigenvectors  | Not diagonalizable.                            |

**Exam memory.** Repeated eigenvalues are not the problem by themselves. The issue is whether there are enough independent eigenvectors.

## 6 Powers and Difference Equations

Diagonalization makes powers easy:

$$A^k = S\Lambda^k S^{-1}.$$

For a difference equation

$$u_{k+1} = Au_k,$$

the solution is

$$u_k = A^k u_0 = S\Lambda^k S^{-1} u_0.$$

If

$$u_0 = c_1 x_1 + \cdots + c_n x_n,$$

then

$$u_k = c_1 \lambda_1^k x_1 + \cdots + c_n \lambda_n^k x_n.$$

## 7 Stability

| Eigenvalue behavior                      | Effect on $A^k$                             |
|------------------------------------------|---------------------------------------------|
| All $ \lambda_i  < 1$                    | Decays to zero.                             |
| Some $ \lambda_i  > 1$                   | Blows up in that direction.                 |
| $ \lambda_i  = 1$                        | Borderline; behavior depends on the matrix. |
| One $\lambda = 1$ , rest $ \lambda  < 1$ | Possible steady state.                      |

## 8 Common Mistakes

| Mistake                                            | Correct exam habit                                                             |
|----------------------------------------------------|--------------------------------------------------------------------------------|
| Solving $Ax = 0$ for every eigenvector.            | Solve $(A - \lambda I)x = 0$ .                                                 |
| Forgetting $x \neq 0$ .                            | Eigenvectors are never the zero vector.                                        |
| Assuming $\lambda = 0$ means diagonalizable fails. | It means $A$ is singular; diagonalizability is separate.                       |
| Assuming repeated eigenvalues always fail.         | Count independent eigenvectors.                                                |
| Row-reducing $A$ before finding eigenvalues.       | Use $\det(A - \lambda I) = 0$ .                                                |
| Confusing invertibility and diagonalizability.     | Invertible means no zero eigenvalue; diagonalizable means enough eigenvectors. |

## 9 Final Exam Checklist

### Checklist.

1. Compute  $\det(A - \lambda I)$ .
2. Solve for all eigenvalues  $\lambda$ .
3. Check trace and determinant against the eigenvalues.
4. For each  $\lambda$ , solve  $(A - \lambda I)x = 0$ .
5. Count independent eigenvectors.
6. Diagonalize only if there are  $n$  independent eigenvectors.
7. Form  $S$  from eigenvectors and  $\Lambda$  from eigenvalues.
8. Use  $A^k = S\Lambda^k S^{-1}$  for powers.
9. Use  $|\lambda|$  to decide growth or decay in difference equations.